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PROFESSIONAL SUMMARY

Automation Engineering graduate (First Class Honors) with hands-on experience in computer vision, 3D deep learning, and robotic perception systems. **Best Paper Award winner at IEEE IcoICT 2025** for developing an optimized 3D CNN achieving 99.75% accuracy in real-time infrared gas leak detection. Developed SLAM algorithms optimized for edge deployment on embedded platforms. Proficient in Python and C++ for perception applications with practical experience in camera calibration, sensor integration, and ROS2-based autonomous navigation. Strong foundation in image processing, temporal-spatial feature extraction, and algorithm optimization for resource-constrained robotic environments.

RESEARCH EXPERIENCES / PROFESSIONAL EXPERIENCES

Full Time Machine Learning Researcher – Maintenance Technology Center (MTC) - Non-Destructive Detection Technology Lab

Sep 2025 – Present

- Designed and built efficient data pipelines and workflows using Python and improved data processing efficiency, scalability, and reliability by optimizing pipeline workflow, resulting in faster model training and deployment.
- Developing advanced machine learning solutions for non-destructive testing applications under Prof. Cherdpong Jomdecha, focusing on acoustic emission sensor data analysis and anomaly detection.
- Implementing unsupervised clustering and anomaly detection algorithms for pattern recognition in complex time-series sensor data from industrial pipe inspection systems
- Contributing to innovative maintenance technology solutions for industrial applications.

Computer Vision (Student) Researcher – OPTIMISED 3D-CNN FOR REALTIME INFRARED NATURAL GAS LEAK CLASSIFICATION

Jun 2024 – Sep 2025 | Bangkok, TH

- Developed optimized 3D CNN architecture for real-time infrared video classification achieving 99.75% accuracy in methane gas leak detection, processing temporal-spatial features from thermal camera streams.
- Implemented classical image processing preprocessing pipeline including Gaussian Mixture Models (GMM), Running Average, and adaptive background subtraction for robust foreground segmentation in varying lighting conditions
- Optimized model for edge deployment through input resolution scaling (half/quarter size variants) and quantization techniques, maintaining 99%+ accuracy while reducing inference time by 60%.
- Co-Winner of Best Paper Award at 13th International Conference on Information and Communication Technology Bandung (Hybrid) - IcoICT 2025.

Junior Machine Learning Intern / Large Language Model Intern – HoneyHunt.AI Dec 2024 – Feb 2025 | Bangkok, TH

- Enhanced automated resume parsing and intelligent job matching systems by integrating AI APIs (advanced LLM APIs), improving accuracy and scalability.
- Improved algorithm scalability and matching accuracy for production-level deployment.
- Contributed to product development in AI-powered recruitment technology.
- Explored Generative AI techniques to design agentic workflows for automated data processing and decision-making, leveraging LLMs and prompt engineering to optimize AI-driven recruitment solutions.
- Implemented API management best practices including versioning, rate limiting, and authentication using API gateways to support reliable deployment of GenAI workflows in production environments.

Machine Learning Intern – Maintenance Technology Center (MTC) - Non-Destructive Detection Technology Lab

Jun 2025 – Aug 2025 | Bangkok, TH

- Independently designed and implemented unsupervised machine learning solutions for GRP pipe inspection using acoustic emission sensor data, demonstrating strong problem-solving and analytical skills in identifying patterns from complex, real-world infrastructure data on Linux based WSL2 platform with version control via Git.
- Won First Runner-Up Award in "Smart Infrastructure for Smart City" competition by Metropolitan Waterworks Authority.
- Listed in KMUTT Hall of Fame for outstanding achievement in Smart Infrastructure development.
- Applied AI/ML techniques to solve real-world infrastructure challenges.

Perception (Student) Research Engineer – CODIA Lab, KMUTT & Panasonic Singapore

Feb 2024 – Jul 2024

- Developed SLAM Sparse Bundle Adjustment algorithms in C.
- Optimized SLAM Sparse Bundle Adjustment algorithms for edge devices, improving computational efficiency and enabling real-time performance in resource-constrained environments.

- Developed and optimized SLAM Sparse Bundle Adjustment algorithms in C for visual odometry applications for real-time edge deployment on ARM-based embedded platforms.

Student Researcher – COSIN Lab, INC Department

Feb 2024 – Jul 2024 | Bangkok, TH

- Developed ROS2-based autonomous navigation systems integrating IMU and camera sensor fusion for competitive robotics applications
- Implemented micro-ROS framework on microcontrollers for real-time IMU data acquisition and sensor synchronization, achieving sub-10ms latency for control loop feedback
- Performed sensor calibration including IMU bias estimation and camera extrinsic alignment to improve localization accuracy in dynamic environments

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT) – Bangkok, Thailand

Jan 2025

Bachelor of Engineering in Automation Engineering

GPAX: 3.66/4.00; Thesis: Natural Gas Methane Leak Detection Using an Infrared Camera; Department: Control System and Instrumentation (INC)

LEADERSHIP & ACTIVITIES

Mentor – FIRST GLOBAL CHALLENGE 2019, Dubai, UAE

Jun 2019 – Oct 2019

- Provided technical guidance and strategic mentorship to Myanmar national robotics team, fostering knowledge sharing through structured training sessions and collaborative problem-solving.
- Led team to win Judges Award at international robotics competition.
- Fostered STEM education and international collaboration in robotics.

Robotics Researcher and Founder – Laboratory of Robotics and Artificial Intelligence (L.O.R.A), Yangon, Myanmar

Dec 2019 – Jun 2021

- Founded and led comprehensive robotics research laboratory focusing on practical applications.
- Spearheaded development of autonomous disinfection robots during COVID-19 pandemic in Myanmar.
- Managed multiple concurrent research initiatives and mentored junior researchers.
- Managed cross-functional communication between engineering teams, public health stakeholders, and external partners to align technical development with societal needs during the pandemic.

AWARDS & RECOGNITION

Best Paper Award – 13th IEEE International Conference on ICT (IcoICT 2025)

Jul 2025

First Runner-Up – Smart Infrastructure for Smart City Competition, Metropolitan Waterworks Authority

Jul 2025

KMUTT Hall of Fame – Outstanding Achievement in Smart Infrastructure Development

Aug 2025

Judges Award – FIRST GLOBAL Challenge 2019 (Team Mentor, Myanmar National Team)

Oct 2019

TECHNICAL SKILLS

Perception & Vision: OpenCV , SLAM (Sparse Bundle Adjustment), Computer Vision, Image Processing , Background Subtraction, Feature Extraction, Camera Calibration

Robotics: ROS2, Robot Kinematics and COnTrol, Sensor Fusion, micro-ROS, Autonomous Naivgation, Gazebo Simulation

Programming Language: Python, C++, C & Embedded C, Bash Scripting

ML/DL: PyTorch, Tensorflow, 3D CNN, Object Detection (YOLO), Segmentation, Model Quantization

Tools: Git, Linux, WSL2

LANGUAGES

English (Fluent (TOEFL 84 - Writing: 25, Speaking: 26 & Extensive technical documentation and international collaboration experience),
Burmese (Native)